Sprint 3 Retrospective

Sprint Duration: Sprint 3
Date Discussed: 10-05-2026

Overview

Sprint 3 focused heavily on implementing major system functionality, integrating payments, refining the user experience, improving vendor and admin workflows, and preparing the system for the final presentation. Throughout the sprint, the team balanced technical implementation, documentation, testing, deployment coordination, and continuous refinement while also managing academic workloads from other university modules.

The sprint introduced some of the most technically complex features completed so far in the project, including:

- Payment-system integration
- Refund functionality
- Vendor analytics
- Order timestamps
- Admin approval workflows
- Vendor/customer profile management
- Browse-page refinements
- Cancel-order functionality

The sprint also highlighted the importance of external feedback, proper issue tracking, deployment coordination, and collaborative debugging.

What Went Well

1. Team Collaboration and Communication

- Team communication remained consistent throughout the sprint through regular online meetings and daily scrum discussions
- Members coordinated dependent pull requests effectively to reduce merge conflicts

- Team members frequently assisted one another with debugging, testing, deployment issues, and Git workflows
- Work was distributed relatively evenly across frontend, backend, analytics, testing, and documentation tasks
- The team maintained a positive and collaborative atmosphere despite growing workload pressure

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2. Significant Functional Progress

The team successfully implemented several major Sprint 3 features:

Completed Core Features

- Payment-system integration
- Refund handling for cancelled pending orders
- Vendor analytics dashboard
- Cancel-order functionality
- Vendor and customer profile pages
- Admin menu-item approval workflow

Improvements Added During Sprint

- Vendor location visibility
- Improved filtering and sorting functionality
- Better dashboard organisation
- Status-based order restrictions
- More structured order management

The team agreed Sprint 3 introduced the largest amount of functional growth compared to previous sprints.

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3. External Feedback Improved the System

One of the most valuable aspects of Sprint 3 was receiving strong external feedback regarding the usability of the platform.

Feedback highlighted issues such as:

- Browse page clutter
- Poor vendor visibility
- Weak organisation of vendor items
- Navigation concerns
- User-experience inconsistencies

Rather than ignoring the criticism, the team used the feedback constructively to:

- Improve filtering systems
- Add vendor location visibility
- Refine navigation flow
- Improve browse-page structure
- Reconsider dashboard layouts
- Add additional UX refinements

The team recognised that developers often become too familiar with their own systems and external perspectives were extremely useful for identifying usability issues.

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4. Payment-System Integration Success

The payment-system implementation was one of the most technically challenging parts of the sprint.

The team successfully:

- Integrated sandbox payment processing
- Added refund support for cancelled pending orders
- Implemented payout validation
- Added payment confirmation flows
- Tested vendor payout handling
- Restricted checkout functionality to customers only

The payment workflow became one of the strongest demonstration features for the Sprint 3 presentation.

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5. Improved Testing and Refinement Mindset

Throughout Sprint 3, the team became more focused on:

- Boundary testing
- Validation handling
- Error prevention
- Presentation-quality UI refinements
- Bug tracking
- Issue management

The team began actively identifying smaller UX and system issues before tutors or external users could discover them.

Challenges Faced

1. Heavy Academic Workload and Time Pressure

The biggest challenge throughout Sprint 3 was balancing project work alongside other university commitments.

The team frequently mentioned:

- Multiple assignments due simultaneously
- Tests and practicals from other modules
- Limited free time during April and May

Although the sprint was productive, members felt the sprint duration coincided with one of the busiest academic periods of the semester.

Team Reflection

Several members stated that:

- The sprint felt very long
- Pressure increased significantly toward the final week
- Earlier project timelines may have reduced stress levels

Despite this, the team still completed most planned functionality successfully.

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2. Merge Conflicts and Shared File Coordination

To reduce conflicts, the team:

- Delayed certain tasks until PRs were merged
- Coordinated file ownership during meetings
- Worked sequentially on dependent tasks
- Performed live PR reviews together

This highlighted the importance of stronger issue assignment and file coordination in future sprints.

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3. Deployment and Environment Challenges

The payment system introduced new deployment-related challenges.

Issues included:

- Payment functionality only working on deployed domains
- Vercel access configuration problems
- Vendor payout validation issues
- Sandbox testing limitations
- API and environment configuration updates

The team spent considerable time:

- Testing deployed environments
- Updating deployment diagrams
- Cleaning invalid vendor data
- Fixing environment-specific bugs

Despite the challenges, the team gained valuable experience working with deployment pipelines and real-world payment-system constraints.

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4. Documentation and Diagram Updates

As the system became more complex, maintaining updated documentation and UML diagrams became increasingly difficult.

Challenges included:

- Updating deployment diagrams after payment integration
- Revising component diagrams

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5. Inconsistent Issue Tracking Early in the Sprint

The team realised midway through Sprint 3 that insufficient GitHub issue tracking created confusion around:

- Task ownership
- Refinement responsibilities
- Bug management
- Remaining sprint work

This sometimes caused:

- Duplicate work
- Delayed refinements
- Uncertainty around feature ownership

Toward the end of the sprint, the team improved issue tracking significantly by:

- Creating more refinement issues
- Labelling bug-related tasks
- Tracking UI improvements more clearly

Lessons Learned

1. External Feedback Is Extremely Valuable

The team learned that honest external feedback can significantly improve a system.

Although some criticism initially felt harsh, it ultimately:

- Improved usability
- Strengthened presentation quality
- Revealed issues developers overlooked
- Encouraged more user-focused thinking

The team agreed future sprints should involve earlier and more frequent external testing.

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2. Continuous Refinement Matters

Sprint 3 reinforced that good software development is not only about functionality.

Important refinement areas included:

- Navigation consistency
- Better layouts
- Clearer dashboards
- Validation handling
- User guidance
- Empty-state displays
- Cleaner UI structure

The team recognised that smaller refinements can greatly improve the overall user experience.

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3. Strong Issue Tracking Prevents Confusion

The sprint demonstrated how important structured issue management is for team coordination.

The team learned to:

- Create issues earlier
- Assign responsibilities clearly
- Separate refinements from major functionality
- Track bugs consistently
- Coordinate shared files more carefully

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4. Real-World Systems Introduce Unexpected Complexity

Integrating payments and refunds showed the team that real-world systems often involve:

- Deployment limitations
- Validation requirements

- Third-party API restrictions
- Security considerations
- Data consistency challenges

The sprint gave the team practical exposure to handling production-style workflows rather than purely academic functionality.

What Could Be Improved

Areas for Future Improvement

- Earlier refinement and testing phases
- Better issue tracking from sprint start
- More consistent documentation updates
- Improved branch/file coordination
- Earlier deployment testing

Overall Sprint Reflection

Sprint 3 was the most demanding sprint so far, both technically and academically. However, it was also the sprint where the team demonstrated the strongest collaboration, adaptability, and technical growth.

Despite workload pressure, deployment challenges, merge conflicts, and evolving requirements, the team successfully delivered a large number of meaningful features while also improving the overall usability and professionalism of the platform.

The sprint reinforced the importance of:

- Communication
- Adaptability
- User feedback
- Incremental refinement
- Testing
- Collaboration

Overall, the team considered Sprint 3 successful and felt significantly more confident about the final presentation and future Sprint 4 enhancements.